

Harsh Vassaram

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SUMMARY

Computer Science Graduate from Arizona State University (GPA: 3.74) with hands-on experience in AI-powered applications and Retrieval-Augmented Generation (RAG) systems. Passionate about software engineering and emerging AI technologies, with a strong foundation in Python, distributed systems, and frontend development. Self-driven tech enthusiast who continuously explores Large Language Models (LLMs), context engineering, vector databases, data curation, and AI app development beyond the classroom. Experienced in transforming research concepts into practical software solutions through academic research, capstone projects, and independent learning.

EDUCATION

Bachelor of Science in Computer Science May 2026
Arizona State University - Tempe, AZ GPA: 3.74

Associate of Science in Computer Science Dec 2023
Mesa Community College - Mesa, AZ

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#, Java, Typescript, JavaScript
- **Frameworks & Backend:** React, Vite, .NET, REST API integration
- **System & Cloud:** Linux, Distributed Systems, Microsoft Azure, GitHub Actions, CI/CD
- **AI & Machine Learning:** RAG Systems, LLMs, Vector Databases, Context Engineering, AI workflow optimization
- **Databases & Tools:** Git, VS Code/Cursor, Figma, PostgreSQL

PROJECTS

Fulton Undergraduate Research Initiative - Chunking Strategies for RAG [\(link\)](#) Spring 2026

- Engineered and benchmarked different chunking strategies (fixed, semantic, hybrid) in RAG pipelines.
- Analyzed domain-specific data sources for optimized data segmentation, reducing LLM hallucinations.
- Evaluated computational cost vs. response accuracy trade-offs to achieve high-fidelity generation.

Viking Roots - AI-Powered Heritage Storytelling Platform (Capstone Project) Fall 2025 - Spring 2026

- Redesigned and implemented production-ready frontend using **React, Vite, and Typescript**, translating Figma prototypes into a deployed platform. Improved user workflow navigation by **20%**.
- Refined Gemini API-powered AI questionnaire through prompt engineering and iterative testing.

Predictive Machine Learning Modeling for Bank Customer Retention Fall 2025

- Developed an Artificial Neural Network (ANN) with **90% prediction accuracy** for customer subscriptions.
- Designed a preprocessing pipeline for **45000+ records**, including feature engineering and one-hot encoding across **17 variables**. Conducted a bias report on the call duration metrics to improve fairness and business impact.

WORK EXPERIENCE

Engineering Dean's Office Operations Assistant Mar 2024 - May 2026

Ira A. Fulton Schools of Engineering, Arizona State University, Tempe, AZ

- Aided in data analysis using Excel Power Query, which led to an increase in faculty reporting efficiency by **30%**.

Front Desk Support May 2022 - Dec 2023

International Education Office, Mesa Community College, Mesa, AZ

- Collaborated with Keystone Education on a data-driven project to increase student enrollment by **20%**.

LEADERSHIP EXPERIENCE

Fulton Academic Integrity Matters - Student Representative Mar 2025 - May 2026

Ira A. Fulton Schools of Engineering, Arizona State University

- Represented the student perspective in academic integrity cases, recommending outcomes.

LANGUAGES: English (Fluent), Portuguese (Fluent), Gujarati (Native), Hindi (Fluent)